Sabi: Validation Evidence

Team Sabi. NAIC 2026, PS3 Sectoral Fine-Tuning (Education)

1. Benchmark A: 522 held-out single questions

174 new WAEC-style questions per subject, never used in training. Scored by code, not by an AI judge.

Measure	Base N-ATLAS	Sabi
Gives the answer away in the first reply (lower is better)	32.2%	2.1%
Asks the student a guiding question	15.3%	100.0%
Uses a Nigerian everyday reference	7.3%	64.0%
Average reply length (words)	112	48

Paired McNemar test on answer leaks: 159 questions where only base N-ATLAS leaked, 2 where only Sabi did, p = 8.9 x 10^-45.

Subject (174 each)	Leaks: base to Sabi	Asks a question	Nigerian reference	McNemar p
Accounting	19.5% to 0.0%	9.8% to 100%	2.3% to 66.7%	1.2e-10
Literature	51.1% to 5.2%	20.1% to 100%	15.5% to 50.6%	1.7e-24
Physics	25.9% to 1.1%	16.1% to 100%	4.0% to 74.7%	1.6e-11

2. Benchmark B: 120 held-out full conversations

40 dialogues per subject, replayed against both models and labelled turn by turn by a separate model (gpt-oss-120b) that did not know which model wrote each transcript.

Measure	Base N-ATLAS	Sabi
Withholds the answer appropriately	40.0%	95.8%
Steps in correctly when the student stays stuck (24 cases)	45.8%	95.8%
Student reaches the correct understanding	87.5%	100.0%
Tutor turns using a Nigerian analogy	5.6%	26.3%

Subject (40 each)	Withholds: base to Sabi	Correct understanding: base to Sabi
Accounting	27.5% to 95.0%	77.5% to 100%
Literature	60.0% to 95.0%	97.5% to 100%
Physics	32.5% to 97.5%	87.5% to 100%

3. Live test of the deployed tutor

On 12 October 2026 we chatted with the live fine-tuned model through the public app in a real browser: 9 messages across Physics, Literature and Accounting, written the way students write. All 9 got a reply (4 to 17 seconds each), none gave the answer away on a first ask, and Sabi used Nigerian examples such as a danfo crossing Third Mainland Bridge and a trader buying bags of rice. Two weak spots we saw: it did not confirm a correct number inside a word problem ("so a = 4?" got "Almost there"), and it answered one direct yes or no question about a metaphor outright.

4. Checking the game's 1,200 questions

All 1,080 practice questions we wrote were answered by a separate model (gpt-oss-120b) that did not see our answer key. It disagreed on 13 (1.2 percent). We reviewed all 13 by hand and our key was correct each time; they were partnership and subscription accounting items where general models often slip.

5. Automated tests

Area	**Checks passing**
10-task stops, replays and fuel	31
Full game flow, offline mode, mobile layout	35
Class levels (Primary, JSS, SS)	25
Security, privacy and rate limits	47
Names, school list, feedback	37
Ask Sabi subjects and read-aloud	20
Calculator coach	18
Exam mode and exam countdown	32
Avatar sounds and mascot	18

6. What we have not done yet

We have not yet run sessions with real students. A pilot plan, parent note and observation sheet are ready in docs/pilot_protocol.md, and the app already collects anonymous one-tap feedback after each stop.